

Data Card: SLM for Agriculture and Climate

Dataset supporting the generation track of the agriculture-slm-challenge (fine-tuning a small language model to answer smallholder farmer questions)

Input vs. Output

What are users inputting?

A short natural-language question from a smallholder farmer or extension worker (e.g. "Mouldy groundnuts after harvest — risk?"), accompanied by structured metadata: the question's topic (one of 8: crop_diseases, pests, soil_health, climate_adaptation, water_management, fertiliser, livestock, post_harvest), crop (maize, beans, cassava, sorghum, groundnuts, livestock, or general), and agro_zone (semi_arid, sub_humid, or highland).

What are the outputs the model predicts or generates?

A single concise generated sentence (typically 8–14 words) that answers the question, grounded in the content of a matched extension factsheet. This is a text-generation output, not a fixed label.

Classification Problem

What is the classification task, if any?

The primary task is text generation, not classification. However, an implicit categorization step underlies it: each question's (topic, crop, agro_zone) metadata maps it to exactly one of 24 source documents — verified during EDA to be a reliable one-to-one mapping for all 12 test questions, with one ambiguous combination (livestock / livestock / sub_humid, spanning 2 documents) observed only in the training data.

What categories or classes are you working with, and why?

24 document classes (one per extension factsheet), organized along three metadata axes: 8 topics, 7 crop categories, and 3 agro-ecological zones. These reflect the practical concerns of smallholder farming in Africa (crop-specific agronomy, livestock care, and zone-specific climate and water constraints).

Labeling

Where do your labels come from?

Both the 24 source factsheets and the 45 training reference answers are recorded as synthetic (origin = "synthetic" for all 24 documents, confirmed via inspection of documents.csv). No document carries a source_url, and no field indicates review by a domain expert, farmer, or extension worker. This is a limitation the team is transparent about: the dataset was constructed for benchmark purposes and does not yet reflect vetted agronomic expertise or community input.

How will you check for consistency and accuracy?

During EDA, we verified structural consistency: every training question's (topic, crop, agro_zone) metadata matched its linked document's own metadata with zero mismatches, and reference answers were confirmed to be non-verbatim, tightly grounded paraphrases of specific clauses within their source document (not hallucinated content unrelated to the factsheet). We have not independently verified the underlying agronomic accuracy of the synthetic content itself, since this was outside the scope of the benchmark and the team has no domain-expert reviewer on hand.

Data Creation

How will you collect your data?

No new data was collected by the team; the dataset (documents.csv, train_qa.csv, test_questions.csv) was provided in full by the competition organizers.

Will it come from direct sources, or will you synthesize data in low-resource settings? Per the documents.csv metadata, all 24 factsheets are synthetic rather than sourced from direct interviews, field recordings, or real extension publications. It also means the dataset does not capture authentic farmer phrasing, dialectal variation, or locally specific practices that real interviews would surface.

Web Collection Ethics

Do you plan to source data from the web?

No. The provided dataset was not scraped from the web — all 24 documents show source_url as null and origin as "synthetic".

If yes, which sites or platforms? What tools will you use? Have you addressed consent, copyright, and privacy? Not applicable to data sourcing, since no web collection occurred. On copyright: all 24 documents are licensed CC0-1.0 (public domain dedication), so there are no copyright restrictions on their use, redistribution, or modification for this project. No personal or personally identifiable data (e.g. named farmers, specific locations) appears in the corpus, so no consent or privacy concerns arise from the dataset itself.

Bias and Representation

Who might be left out of your dataset?

Coverage is narrow by construction: only 7 crop categories (maize, beans, cassava, sorghum, groundnuts, livestock, general) and 3 agro-ecological zones (semi_arid, sub_humid, highland) are represented. Farmers growing crops outside this list (e.g. coffee, tea, bananas, horticultural crops) or working in zones such as arid or coastal/humid regions are not served by this dataset as it stands. Because the source material is synthetic rather than drawn from real community input, the phrasing, concerns, and practices reflected may also not match how farmers in any specific locality actually describe their problems.

How will you ensure the dataset reflects the diversity of the community and includes underrepresented voices? At this stage, the team has not added community-sourced data on top of the provided synthetic corpus, so this dataset alone should not be presented as representative of any real farming community. If this work continues beyond the benchmark, a clear next step would be validating factsheet content and question phrasing with actual extension workers or farmer groups across a wider range of crops and zones before any real-world deployment.

Values and Community Alignment

The team's guiding value is accuracy and trustworthiness over generic AI advice — the concern that general-purpose LLMs can hallucinate specific rates, timings, or practices that carry real agronomic risk if acted on incorrectly. The dataset design reflects this by tightly coupling every question to a single verified source document via topic/crop/agro_zone metadata, and by scoring generation against short, specific reference answers (via Levenshtein distance) rather than open-ended free text — rewarding answers that stay close to a

grounded, specific source rather than plausible-sounding generalities.

How will your community benefit, and how will you ensure accountability to them?

In its current synthetic-data form, this dataset is a benchmark and prototype, not a deployed community resource, so direct farmer benefit is not yet realized. Accountability going forward would mean being explicit with any future users that the underlying factsheets are synthetic and unvetted, avoiding presenting model outputs as verified agronomic advice until real-world validation with domain experts and the farming communities the tool is meant to serve has taken place.